

PXR0 single-file physical runtime - exhaustive active specification v1

PX8 authority 9ba11aeb4f88d6707cfba1afdb5f6dce3e380b9f; PXR0 development only

Canonical source SHA-256: f6989555f5a43dff91b39a5c7f79038168f39142fdbecca7e5e40938a72785cb

Physical state: cells, arrows, queued spikes, local time, generation/live state, resistance, eligibility and activation.

Local rules: threshold/refractory firing, Drive traversal, qualified Modulation, decay, pressure, deallocation and nearby proposal.

Continuous execution: enter appends a physical spike; propagate orders the queue and advances until natural quiescence.

External crossings: RunResult exposes ordinary inter-region crossings, bounded work, quiescence and resident bytes.

All active types/state (13)

CellId - tuple index naming one cell.

ArrowId - tuple index naming one arrow.

CellSpec - physical_id, position, region, threshold, resistance bootstrap fields.

TransmissionMode - Drive or Modulatory physical transmission.

ArrowSpec - from, to, delay, phase, coupling, resistance, mode bootstrap fields.

SpikeInput - arrival_tick, phase, origin_physical, target, impulse boundary fields.

Cell - physical_id, position, region, threshold, activation, last tick, refractory limit, generation, resistance, live.

Arrow - from/to, delay, phase, coupling, source generation, generation, resistance, live, eligibility, mode.

Spike - arrival tick, phase, physical origin, target/generation, impulse, serial, optional arrow/generation.

Crossing - tick, physical endpoints, endpoint regions and impulse.

Work - total plus Drive, Modulation, local-update, proposal and deallocation counts.

RunResult - crossings, work, natural-quiescence flag and resident bytes.

PlasticSubstrate - cell/arrow/spike vectors, tick, next serial and pressure tick.

All active functions/methods (15)

Work::total - returns the accumulated bounded-work count without mutation.

PlasticSubstrate::new - creates empty physical state.

PlasticSubstrate::add_cell - validates and appends one generic cell.

PlasticSubstrate::add_arrow - validates endpoints/delay and appends one generic arrow.

PlasticSubstrate::enter - validates time/target and appends one external spike.

PlasticSubstrate::advance_time - applies local elapsed-time dynamics with an empty queue.

PlasticSubstrate::propagate - delivers the ordered queue to natural quiescence and returns crossings/work.

PlasticSubstrate::apply_modulatory_return - strengthens only live eligible arrows from the reached cell.

PlasticSubstrate::elapse_to - applies ordinary pressure, expiry pressure and cell decay.

PlasticSubstrate::propose_local_arrows - adds deterministic nearby Drive arrows after external firing.

PlasticSubstrate::decay_cell - moves activation toward zero over elapsed local time.

PlasticSubstrate::require_cell - enforces substrate-local cell identity.

PlasticSubstrate::spike_order - orders arrival tick, phase, origin, target identity and serial.

PlasticSubstrate::resident_bytes - reports cell plus arrow storage for the registered bound.

pressure_arrow - reduces resistance and retires a zero-resistance arrow by generation.